York University

Lassonde School of Engineering

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Assignment Title

Code: Class

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Instructor:
Dr. Professor

Due date

EXAMPLE

Abstract

Type abstract here.

EXAMPLE

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EXAMPLE

Todo list

This is a margin to do note. 20
This is an inline to do note. 20

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EXAMPLE

1 Conditionals

1.1 xifthen

The `xifthen` package extends LaTeX's conditional logic with `\ifthenelse` and `\isempty`. It is used throughout this template to conditionally load packages and apply settings based on the metadata macros defined at the top of `main.tex`. It produces no visible output on its own.

No partners defined (default).

2 Colors

2.1 xcolor

The `xcolor` package provides extended color support. The template defines `facultycolour` as the primary brand color, automatically selected based on `\metaFaculty` and `\metaFacultyColour` in the metadata. It is used throughout headings, boxes, and rules.

Table 1: Faculty colour and tints.

Color Name	Hex	Preview
`facultycolour`	100% tint	
`facultycolour!70`	70% tint	
`facultycolour!40`	40% tint	

Inline color: This text is in the brand colour.

3 Page Geometry

3.1 geometry

The `geometry` package sets the page margins. This template uses 2.54 cm (1 inch) margins on all sides with `headheight=14.9pt` to accommodate the header. These are set in `main.tex` and produce no visible output here.

3.2 setspace

The `setspace` package controls line spacing. The template supports `single`, `onehalf`, and `double` spacing, set via `\metaLineSpacing` in the metadata. The current document uses `onehalf` spacing.

EXAMPLE

4 Encoding & Language

4.1 `inputenc`

The `inputenc` package with `utf8` option allows the source file to contain UTF-8 characters directly, such as accented letters: é, ü, ñ, ç.

4.2 `fontenc`

The `fontenc` package with `T1` option enables 8-bit font encoding, which improves glyph rendering and hyphenation for non-ASCII characters. It produces no visible output on its own.

4.3 `babel`

The `babel` package with `english` applies English hyphenation rules. For example, the word "internationalization" will be hyphenated correctly across line breaks.

5 Font Selection

5.1 `plex-serif` and `\seriftext`

The `plex-serif` package loads IBM Plex Serif, York University's brand serif font. Use `\seriftext{}` to apply it to specific text:

This text is rendered in IBM Plex Serif (York University brand serif font).

5.2 `plex-sans` and `\sansseriftext`

The `plex-sans` package loads IBM Plex Sans. Use `\sansseriftext{}` to apply it to specific text:

This text is rendered in IBM Plex Sans (York University brand sans-serif font).

5.3 Default Font

The default document font is controlled by `\metaDefaultFont` in the metadata. The current setting is `sans`. Changing it to `serif` switches the entire document to IBM Plex Serif.

6 Spacing Around Headings

EXAMPLE

6.1 `titlesec`

The `titlesec` package customises heading fonts, sizes, colors, and spacing. All headings in this document use `facultycolour` and bold formatting. The spacing above and below each heading level is controlled by `\titlespacing*`.

6.1.1 This is a `\subsubsection`

Subsubsections use `\normalsize\bfseries` in `facultycolour`.

6.1.2 Abstract environment

The `abstract` environment is redefined in `yorku.sty` using `titlesec` to render as an unnumbered section heading rather than the default centred italic block. The abstract at the top of this document demonstrates the result.

7 Math

7.1 `amsmath`, `amssymb`, `amsfonts`

The AMS packages provide math environments and symbols:

$$\int_0^\infty e^{-x^2}\,dx = \frac{\sqrt{\pi}}{2} \tag{1}$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6} \tag{2}$$

7.2 `mathtools`

The `mathtools` package extends `amsmath` with additional tools such as `\coloneqq`:

$$f(x) \coloneqq x^2 + 2x + 1 \tag{3}$$

Matrix with `pmatrix`:

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \tag{4}$$

7.3 `bm`

The `bm` package provides bold math symbols via `\bm{}`:

$$\boldsymbol{F} = m\boldsymbol{a} \tag{5}$$

EXAMPLE

7.4 siunitx

The siunitx package formats SI units consistently:

Table 2: SI unit examples using siunitx.

Command	Output
`\SI{9.81}{\metre\per\second\squared}`	9.81 m/s^2
`\SI{3e8}{\metre\per\second}`	3×10^8 m/s
`\SI{100}{\kilo\gram}`	100 kg
`\si{\ohm}`	Ω

7.5 mhchem

The mhchem package typesets chemical formulas and equations via `\ce{}`:
Inline: $\mathrm{H_2O}$, $\mathrm{Na^+}$, $\mathrm{SO_4^{2-}}$

$$\mathrm{2\,H_2 + O_2 \longrightarrow 2\,H_2O} \tag{6}$$

$$\mathrm{A + B \rightleftharpoons C + D} \tag{7}$$

$$\mathrm{A \xrightarrow{\Delta} B} \tag{8}$$

$$\mathrm{^{235}_{92}U} \tag{9}$$

8 Theorem Environments

8.1 amsthm

The amsthm package provides styled theorem-like environments. This template defines three styles: plain, definition, and remark, and a custom proof environment.

8.1.1 Plain — Theorem, Lemma, Proposition, Corollary, Conjecture, Axiom, Proof

Theorem 1: *For any right triangle with legs a and b and hypotenuse c: $a^2 + b^2 = c^2$.*

Proof: Drop a perpendicular from the right angle to the hypotenuse and apply similarity of the resulting triangles. ■

Note that `\qed` is redefined in yorku.sty to render as a solid facultycolour square (■) rather than the default hollow box.

Lemma 1: *If n is an even integer, then n^2 is also even.*

EXAMPLE

Proposition 1: *The sum of two rational numbers is rational.*

Corollary 1: *If p is prime and $p > 2$, then p is odd.*

Conjecture 1: *Every even integer greater than 2 can be expressed as the sum of two primes.*

Axiom 1: *Through any two distinct points there exists exactly one straight line.*

8.1.2 Definition — Definition, Example, Criterion

Definition 1: A **vector space** over a field $\mathbb{F}$ is a set V equipped with addition and scalar multiplication satisfying the vector space axioms.

Example 1: The set $\mathbb{R}^n$ with standard addition and scalar multiplication is a vector space over $\mathbb{R}$.

Criterion 1: A system is stable if and only if all eigenvalues of its state matrix have strictly negative real parts.

8.1.3 Remark — Remark, Note

Remark 1: The zero vector is unique in any vector space.

Note 1: All theorem environments are automatically numbered by section when `\metaNumberTheorems` is set to `true`.

9 Graphics, Figures & Tables

9.1 graphicx

The `graphicx` package provides `\includegraphics`. Images are searched in `figures/` and `assets/` automatically.

Figure 1: The faculty logo included with `\includegraphics` (this figure requires that a file pathway is in the `metaLogoPath` metadata).

9.2 booktabs

The `booktabs` package provides professional horizontal rules: `\toprule`, `\midrule`, `\bottomrule`:

EXAMPLE

Table 3: Basic table using `booktabs`.

Parameter	Value	Unit
Resistance	10	Ω
Current	2	A
Voltage	20	V

9.3 `array`

The `array` package extends LaTeX's column formatting options for `tabular` environments, adding new column specifiers and allowing custom column types. It is required by `tabularx` and `multirow` and produces no standalone visible output.

Table 4: Fixed-width centred columns using `array`.

Left column	Right column
Centred text Long cell content wraps	Centred text Also wraps cleanly

9.4 `tabularx`

The `tabularx` package provides auto-width `X` columns that fill the available width:

Table 5: Auto-width table using `tabularx`.

Parameter	Value	Unit	Description
Resistance	10	Ω	Opposition to current flow in a circuit
Current	2	A	Flow of electric charge through a conductor
Voltage	20	V	Electric potential difference between two points

9.5 `multirow`

The `multirow` package allows cells to span multiple rows:

EXAMPLE

Table 6: Table with multi-row cells using `multirow`.

Category	Parameter	Value
Electrical	Resistance	10 Ω
	Current	2 A
Mechanical	Force	50 N
	Velocity	3 m/s

9.6 `longtable`

The `longtable` package allows tables to span multiple pages:

Table 7: A long table that can span multiple pages.

Parameter	Value	Unit
Row 1	1	Ω
Row 2	2	Ω
Row 3	3	Ω
Row 4	4	Ω
Row 5	5	Ω
Row 6	6	Ω
Row 7	7	Ω
Row 8	8	Ω
Row 9	9	Ω
Row 10	10	Ω

9.7 `float`

The `float` package provides the `[H]` placement specifier, which forces a float to appear exactly where it is in the source. All figures and tables in this document use `[H]`.

9.8 `subcaption`

The `subcaption` package provides the `subfigure` environment for side-by-side figures:

EXAMPLE

(a) First sub-figure.

(b) Second sub-figure.

Figure 2: Two sub-figures side by side using `subcaption` (these figures require that a file pathway is in the `metaLogoPath` metadata).

9.9 `caption`

The `caption` package customises caption formatting. All captions in this document use `facultycolour` bold labels with a colon separator, raggedright justification, and `small` font size, as seen in all figures and tables above.

9.10 `pdfpages`

The `pdfpages` package inserts external PDF pages via `\includepdf{filename.pdf}`. Use `pages=-` to include all pages, or `pages={1,3}` to include specific pages. This is useful for attaching datasheets, scanned documents, or supplementary material in the appendix. For a usage example, see Section A.

10 Diagrams & Plots

10.1 `tikz`

The `tikz` package provides programmatic vector graphics. Example of a simple block diagram:

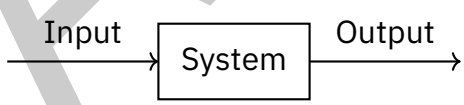

Figure 3: Simple block diagram drawn with `tikz`.

10.2 `pgfplots`

The `pgfplots` package builds data plots on top of TikZ. Example of a simple function plot:

EXAMPLE

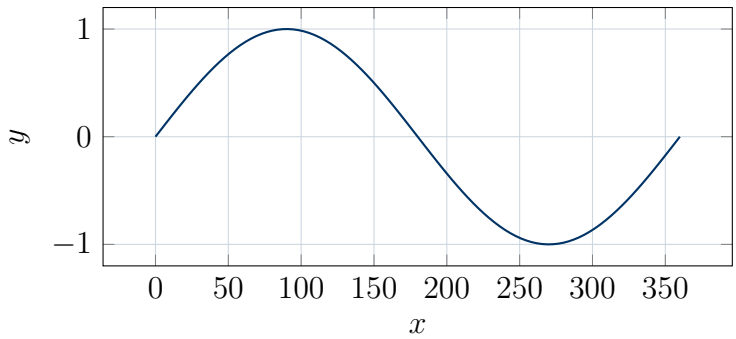

Figure 4: Sine wave plotted with `pgfplots`.

10.3 `circuitikz`

The `circuitikz` package draws circuit schematics. Example of a simple RC circuit:

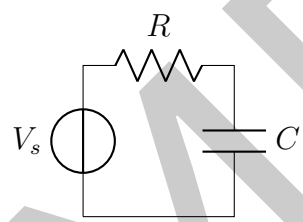

Figure 5: Simple RC circuit drawn with `circuitikz`.

11 Algorithms

11.1 `algorithm` and `algpseudocode`

The `algorithm` package provides a float wrapper for algorithms, giving them captions, labels, and an entry in the list of algorithms. The `algpseudocode` package provides the pseudocode commands. Use `\Require` and `\Ensure` for inputs and outputs, and `\Comment{}` for inline comments.

EXAMPLE

Algorithm 1: Binary Search

Input: Sorted array A of length n, search value $target$

Output: Index of $target$ in A, or -1 if not found

```
function BinarySearch(A, n, target)
    low ← 0
    high ← n − 1
    while low ≤ high do
        mid ← ⌊(low + high)/2⌋                ▷ Compute midpoint
        if A[mid] = target then
            return mid                        ▷ Target found
        else if A[mid] < target then
            low ← mid + 1                     ▷ Search right half
        else
            high ← mid − 1                    ▷ Search left half
        end if
    end while
    return −1                                 ▷ Target not found
end function
```

Algorithm 2: Bubble Sort

Input: Array A of length n

Output: A sorted in ascending order

```
function BubbleSort(A, n)
    for i ← 0 to n − 1 do
        for j ← 0 to n − i − 2 do
            if A[j] > A[j + 1] then
                swap A[j] and A[j + 1]        ▷ Swap out of order elements
            end if
        end for
    end for
end function
```

A reference to an algorithm uses `\cref`: see Algorithm 1 and Algorithm 2.

12 Links & References

EXAMPLE

12.1 hyperref

The hyperref package makes cross-references and URLs clickable. All links use facultycolour. Example URL: https://www.yorku.ca

12.2 cleveref

The cleveref package automatically prepends the reference type. Examples:

- Equation: Eq. (1)
- Figure: Fig. 1
- Sub-figure: Fig. 2a
- Table: Table 3
- Algorithm: Algorithm 1
- Section: Section 14
- Theorem: Theorem 1
- Lemma: Lemma 1
- Proposition: Proposition 1
- Corollary: Corollary 1
- Conjecture: Conjecture 1
- Axiom: Axiom 1
- Definition: Definition 1
- Example: Example 1
- Criterion: Criterion 1
- Remark: Remark 1
- Note: Note 1
- Problem: Problem 1

EXAMPLE

13 Typography

13.1 microtype

The microtype package applies subtle typographic improvements including character protrusion and font expansion. The effect is most visible in justified text with many line breaks. It produces no visually distinct output on its own but improves overall text quality.

13.2 xurl

The xurl package extends hyperref's URL handling by allowing line breaks at more characters, including slashes, dots, hyphens, and underscores. This prevents overfull lines when long URLs appear in the body text. URLs are colored facultycolour via the \url command. Example: https://futurestudents.yorku.ca/program/engineering-intl-development-studies

13.3 indentfirst

The indentfirst package indents the first paragraph of every section, matching the behaviour of the remaining paragraphs.

This is the first paragraph of this subsection — it is indented because indentfirst is loaded.

This is the second paragraph — also indented as normal.

14 Lists

14.1 enumitem

The enumitem package provides fine-grained control over list spacing, labels, and alignment.

Unordered list with reduced spacing:

- First item
- Second item
- Third item

Ordered list with custom label:

1. First item
2. Second item
3. Third item

EXAMPLE

Descriptive list:

`noitemsep`	Removes vertical space between items
`label`	Customises the list label format
`leftmargin`	Controls the left indent of the list

15 Problem & Solution Environments

15.1 `tcolorbox`

The `tcolorbox` package provides the `problem` and `solution` environments. Problems are automatically numbered (by section if `metaNumberProblems` is set to true).

Problem 1: Ohm's Law

Given a resistor $R = 10\,\Omega$ with a voltage $V = 5\,\text{V}$ across it, find the current I.

Solution

Applying Ohm's Law (Eq. (1)):

$$I = \frac{V}{R} = \frac{5}{10} = 0.5\,\text{A} \tag{10}$$

■

Problem 2

A problem without a title is also valid.

Solution

The optional title argument can be omitted. ■

16 Code Listings

16.1 `listings`

The `listings` package provides syntax-highlighted code listings. Five styles are defined: `matlab`, `java`, `python`, `cpp`, and `latex`.

EXAMPLE

16.1.1 MATLAB

```
x = 0:0.1:2*pi;
y = sin(x);
plot(x, y);
xlabel('x'); ylabel('sin(x)');
title('Sine Wave');
```

16.1.2 Java

```
public class HelloWorld {
    public static void main(String[] args) {
        System.out.println("Hello, World!");
    }
}
```

16.1.3 Python

```
import numpy as np
import matplotlib.pyplot as plt

x = np.linspace(0, 2 * np.pi, 100)
y = np.sin(x)
plt.plot(x, y)
plt.show()
```

16.1.4 C++

```
#include <iostream>
using namespace std;

int main() {
    cout << "Hello, World!" << endl;
    return 0;
}
```

16.1.5 LaTeX

```
\begin{equation}
    E = mc^2
    \label{eq:einstein}
\end{equation}
```

```
See \cref{eq:einstein} for Einstein's mass-energy equivalence.
```

17 Draft Tools

17.1 todonotes

The `todonotes` package adds margin and inline notes. Controlled by `\metaShowTodo` in the metadata.

This is a margin to do note.

This is an inline to do note.

17.2 draftwatermark

The `draftwatermark` package overlays a diagonal watermark. Set `\metaWatermarkText` in the metadata to enable it. It is currently enabled.

17.3 lipsum

The `lipsum` package generates dummy text for testing layout:

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

18 Header & Footer

18.1 fancyhdr

The `fancyhdr` package provides custom headers and footers. This template uses it to display the course code and name on the left, the student name on the right, and the page number centred in the footer, on all pages after the title page. The header rule is rendered in `facultycolour`. It produces no additional visible output here beyond what is already visible at the top and bottom of every page.

EXAMPLE

19 Bibliography

19.1 `csquotes`

The `csquotes` package provides context-sensitive quotation marks and is required by `biblatex`. It produces no visible output on its own.

19.2 `biblatex`

The `biblatex` package handles citations and bibliography generation using Biber as the backend. The citation style is controlled by `\metaBibStyle` in the metadata (`ieee` or `apa`). The current style is `ieee`. Add entries to your `.bib` file and cite them with `\cite{}`. Example citation: [1].

EXAMPLE

References

[1] J. Zhang, K. Fu, J. Qiu, X. Xia, H. Nie, and X. Wen, "Vibration characteristics of CFRP sandwich panels with acoustic black hole structures," *Mechanics of Advanced Materials and Structures*, vol. 33, no. 1, p. 2 445 223, Dec. 26, 2024, issn: 1537-6494. doi: 10.1080/15376494.2024.2445223 Accessed: Apr. 26, 2026. [Online]. Available: https://www.tandfonline.com/doi/full/10.1080/15376494.2024.2445223 (cit. on p. 21).

EXAMPLE

Appendix A Supplementary Data

This appendix demonstrates how appendix sections are formatted. Section headings are relabelled as **Appendix A**, **Appendix B**, etc., and figures, tables, and equations are prefixed with the corresponding letter.

Table A.1: Sample supplementary data table.

Parameter	Measured	Unit
Temperature	23.4	°C
Pressure	101.3	kPa
Humidity	48.7	%

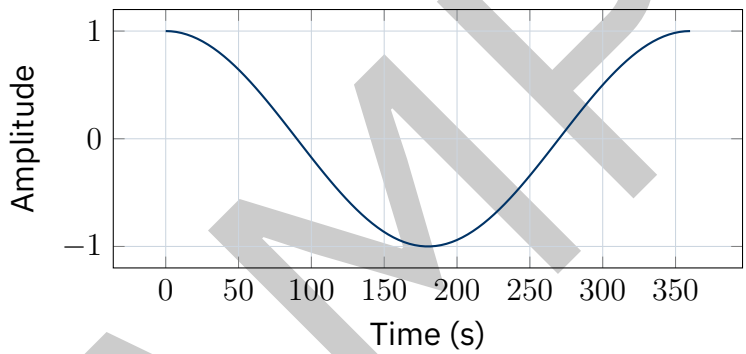

Figure A.1: Sample cosine plot included as supplementary material.

To attach an external PDF (e.g. a datasheet), replace the content above with:

```
\includepdf[pages=-]{filename.pdf}
```

EXAMPLE

Appendix B Notation

This appendix lists the mathematical notation used throughout the document.

Table B.1: Summary of notation.

Symbol	Meaning
$\boldsymbol{F}$	Force vector (N)
m	Mass (kg)
$\boldsymbol{a}$	Acceleration vector (m/s^2)
V	Voltage (V)
I	Current (A)
R	Resistance (Ω)

EXAMPLE